

• Eigenvalues and Eigenvectors

Given a square matrix A , an eigenvector of A is a non-zero vector $\mathbf{v}$ such that when A acts on $\mathbf{v}$, the resulting vector is a scalar multiple of $\mathbf{v}$. Mathematically, this can be expressed as:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Where:

- A is a square matrix (i.e., $n \times n$).
- $\mathbf{v}$ is a non-zero vector called an *eigenvector*.
- λ is a scalar value called an *eigenvalue*.

The equation $A\mathbf{v} = \lambda\mathbf{v}$ means that applying the matrix A to $\mathbf{v}$ simply stretches or compresses $\mathbf{v}$ by the factor λ , but does not change its direction.

Eigenvalues and eigenvectors are essential because they simplify matrix operations, especially when diagonalizing matrices or solving systems of differential equations. Eigenvectors provide insight into the geometric transformations represented by a matrix, while eigenvalues indicate how much vectors are stretched or compressed in the direction of eigenvectors.

To find the eigenvalues and eigenvectors of a matrix A , we follow these steps:

1. Solve the Characteristic Equation

The eigenvalues of a matrix are obtained by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$

Here:

- I is the identity matrix.
- λ is the scalar (the eigenvalue we are solving for).
- $\det(A - \lambda I)$ is the determinant of the matrix $(A - \lambda I)$.

This equation gives us a polynomial (called the *characteristic polynomial*) in λ , and the roots of this polynomial are the eigenvalues of A .

2. Find the Eigenvectors

Once we have the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, we can find the corresponding eigenvectors by substituting each eigenvalue λ_i into the equation:

$$(A - \lambda I)\mathbf{v} = 0$$

and solving for $\mathbf{v}$.

This system of equations typically has infinitely many solutions, so we find a basis for the set of solutions, which will give us the eigenvectors.

Geometric Interpretation of Eigenvalues and Eigenvectors

Eigenvectors represent directions in which a transformation matrix A acts by only stretching or compressing (not rotating or reflecting). Eigenvalues represent the scaling factor by which the eigenvector is stretched or compressed when acted upon by the matrix A .

For example, if $\lambda = 2$, the matrix A stretches the eigenvector by a factor of 2. If $\lambda = -1$, the matrix A reflects the eigenvector and scales it by 1.

• Eigenspaces

The eigenspace corresponding to an eigenvalue λ is the set of all eigenvectors associated with that eigenvalue, along with the zero vector. Formally, the eigenspace of λ is the null space of the matrix $(A - \lambda I)$.

$$\text{Eigenspace of } \lambda = \{\mathbf{v} \mid (A - \lambda I)\mathbf{v} = 0\}$$

The dimension of the eigenspace corresponding to an eigenvalue λ is called the *algebraic multiplicity* of λ . This dimension tells how many linearly independent eigenvectors are associated with λ .

Example: If $\lambda = 0$ is an eigenvalue of A , the eigenspace corresponds to the null space of A , meaning it includes all vectors $\mathbf{v}$ that satisfy $A\mathbf{v} = 0$. This eigenspace is the set of all solutions to the homogeneous system $A\mathbf{v} = 0$.

Problems

a) The set of eigenvalues of the 3×3 real matrix A is $\{0, 1, 2\}$. What is the rank of A ?

1. Understanding the Eigenvalues and Eigenspaces:

The eigenvalues of a matrix are the values λ for which the equation $A\mathbf{v} = \lambda\mathbf{v}$ holds for some non-zero vector $\mathbf{v}$, which is called an **eigenvector**. For each eigenvalue, the set of all eigenvectors corresponding to that eigenvalue, along with the zero vector, forms the **eigenspace** of λ .

Eigenvalue 0:

- If $\lambda = 0$ is an eigenvalue, it means there is a non-trivial **null space**. The null space is the set of vectors $\mathbf{v}$ such that $A\mathbf{v} = 0$. The dimension of this null space is called the **nullity** of the matrix.
- The **eigenspace** corresponding to $\lambda = 0$ is the null space of A .
- Since $\lambda = 0$ is an eigenvalue, the nullity of A is at least 1.

Eigenvalues 1 and 2:

- For $\lambda = 1$ and $\lambda = 2$, each has a corresponding eigenspace, which contains the eigenvectors associated with these eigenvalues.

2. Relationship Between Rank and Nullity The **rank-nullity theorem** states that for any $n \times n$ matrix A :

$$\text{rank}(A) + \text{nullity}(A) = n$$

Here, $n = 3$, since A is a 3×3 matrix. So:

$$\text{rank}(A) + \text{nullity}(A) = 3$$

3. Determine the Rank of A Using the rank-nullity theorem:

- The matrix A has three eigenvalues: 0, 1, 2. The eigenspace corresponding to $\lambda = 0$ gives us the nullity, and the eigenspaces for $\lambda = 1$ and $\lambda = 2$ contribute to the rank.
- If the nullity of A is 1, then the rank of A must be:

$$\text{rank}(A) = 3 - \text{nullity}(A) = 3 - 1 = 2$$

The rank of the matrix A is **2**.

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.